

CLOVER · Sub-Meter Optical Canopy Delineation & Carbon MRV

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1. Approach & Problem Definition

Standard object detection models (e.g., DeepForest, YOLO) output disjoint bounding boxes in image pixel space. In nature, closed-canopy tropical forests and urban woodlands intertwine into a continuous, overlapping canopy. **Directly summing bounding box areas double-counts overlapping foliage by 20% to 40%**, artificially inflating canopy cover and fabricating phantom carbon credits. **Clover's core philosophy** is to bridge the gap between raw computer vision detections and auditor-grade forestry accounting: transform 2D pixel coordinates into projected metric coordinates, topologically dissolve overlapping crowns into a unary union footprint, calibrate for optical area underestimation (Li et al., 2023), and propagate honest stand-level uncertainty compliant with **Verra VM0047 ARR** and **IPCC Tier 2** standards.

2. System Architecture & End-to-End Workflow

Stage 1: Geospatial Ingestion	Stage 2: Tiled Neural Inference	Stage 3: Topological Accounting
- Reads GeoTIFF metadata (CRS, Affine, GSD). - Auto-reprojects to local UTM (EPSG:32645 Kolkata; EPSG:32617 US). - Parses KML/GeoJSON AOI and clips raster.	- Sliding-window tiler (400x400 px, 15% overlap). - DeepForest RetinaNet (ResNet50 backbone). - Custom boundary NMS stitching across seams. - Adaptive multi-scale pyramid (1.0x + 2.0x).	- Fits $\pi/4$ inscribed ellipses to crown boxes. Shapely unary union dissolves all crowns. - Zero overlap double-counting guaranteed. - Purges lake weeds via HSV hydrological filter.

3. Key Technical Decisions & Mathematical Formulations

- Topological Non-Overlapping Dissolution:** Individual crown polygons are merged into a single multi-polygon using geometric unary union: $Canopy\ Cover\ \% = \frac{Area(Union(Crowns))}{Area(AOI)} \times 100$. If two crowns share a 50 m² overlap, it is credited exactly once.
- Empirical Crown Area Bias Correction (+20%):** Based on empirical satellite validation by Li et al. (PNAS Nexus, 2023), optical bounding boxes underestimate true closed-canopy crown area due to edge contrast attenuation. Clover applies a calibrated +20% scaling correction prior to allometric modeling.
- Pantropical Biomass Allometry:** Above-ground biomass (AGB) is derived using pantropical scaling relations (Jucker et al., 2016; Chave et al., 2014): $ln(AGB) = \alpha + \beta \cdot ln(CD) + \gamma \cdot ln(CD)^2 + \epsilon$. AGB is converted to Carbon Stock via IPCC carbon fraction ($C = AGB \times 0.47$) and carbon dioxide equivalents ($tCO_2e = C \times 44/12$).
- Dual-Component Stand Uncertainty Decomposition:** While single-tree random error is large ($\sigma_{rand} \approx \pm 56.5\%$, Jucker 2016), it attenuates over N stems via $\sigma_{rand}/\sqrt{N}$. However, stand-scale biomass models hit an unavoidable systematic allometric and sensor error floor ($\sigma_{sys} \approx 10\text{--}44\%$, Chave 2014; Réjou-Méchain 2017). Clover models both terms honestly: $\sigma_{stand} = \sqrt{[(56.5\%)^2/N + (12.0\%)^2]}$. Stand uncertainty realistically converges to **$\pm 12.5\%$ to $\pm 13.5\%$ (90% CI)**, meeting **Verra VCS Standard v4.5 precision guidelines** (error $\leq 15\%$ receives zero penalty deduction).

Auditability Insight: Naive tools claim 99% accuracy or claim error drops to $\pm 2\%$ across 300 trees. This is statistically impossible in optical forestry due to unobserved tree height and wood density variance. Clover admits this limit and implements the dual-component floor.

4. What Worked & Practical Validations

- Curated Benchmarks Across Diverse Biomes:** Verified across 6 distinct ecological regimes: Kolkata Central Park Banabitan (22.5855°N, 88.4180°E), Kolkata Victoria Memorial Gardens (22.5448°N), Bengaluru Cubbon Park & Lalbagh, Muir Woods Redwood Forest (California), and Sundarbans Mangrove Biosphere (21.9497°N, 88.8995°E). Delineation runs in 0.4–2.5s across all sites.
- Hydrological Lake Weed False-Positive Rejection:** In Kolkata Banabitan and urban lakes, floating water hyacinth (Eichhornia crassipes) triggers false positives in optical detectors. A color-space HSV water-masking filter successfully eliminates aquatic weed false positives without corrupting shoreline overstory trees.
- Sub-Second Local Execution via UV:** Configured pyproject.toml with Astral's uv package manager. Fresh environments resolve and install 120 wheels in ~1.08s, allowing any developer to run the full stack locally via uv run streamlit run app.py without dependency conflicts.
- 24/7 Cloud Zero-Friction Hosting:** Deployed on Streamlit Community Cloud (https://tryclover.streamlit.app). A stranger can open the URL from mobile or desktop, select presets, upload custom GeoTIFFs, inspect tree crowns on satellite imagery, and download CSV/GeoJSON audit files in one click.

5. What Didn't Work & Engineering Dead Ends (Lessons Learned)

- Zero-Shot Segment Anything (SAM / FastSAM) Without Prior Localization:** Directly applying raw SAM across continuous tropical forest patches resulted in severe oversegmentation — dividing single broadleaf trees into 4–10 fragmented sub-clusters (branches and sunlit leaves). **Solution:** DeepForest RetinaNet must act as the primary crown detector, with SAM restricted to intra-box boundary refinement.
- Raw Bounding Box Union:** Dissolving rectangular boxes directly without ellipse fitting over-credited canopy area by ~21.5% at the four diagonal corners of each crown. Inscribing $\pi/4$ geometric ellipses inside detected boxes before running unary union aligned predicted canopy cover with

manual delineations within 90% agreement.

3. Static Ground Sampling Distance (GSD) Assumptions: Applying DeepForest (trained on 10 cm/px airborne NEON data) directly to drone screenshots (2–5 cm/px) or satellite rasters (30–50 cm/px) resulted in catastrophic scale mismatch. Built an adaptive scale normalizer that dynamically rescales imagery to an optimal ~25 cm/px receptive field combined with a dual-resolution (1.0x + 2.0x) feature pyramid.

6. Known Scientific Limitations & Boundary Conditions (Regulatory Transparency)

In strict compliance with Verra VM0047 and IPCC Tier 2 principles, Clover explicitly discloses its operational boundaries in its UI telemetry:

- **Understory Blind Spot:** Sub-meter passive optical sensors capture only the dominant overstory canopy. Suppressed saplings and understory stems shielded beneath emergent crowns are invisible from overhead sensors. Under Verra VM0047 ARR methodologies, optical canopy models must be paired with stratified ground inventory sample plots to quantify understory biomass.
- **Missing 3D Vertical Structure (LiDAR Gap):** 2D optical crown diameter is an allometric proxy for tree volume. Two trees with identical 7-meter crowns can have heights of 14m vs. 28m depending on stand age and competition. Full Tier 3 carbon credit certification requires fusing optical crown polygons with spaceborne LiDAR (GEDI L2A/L2B) or airborne LiDAR canopy height models (CHM).
- **Pantropical Wood Density (ρ) Variance:** Clover assumes regional mean wood density ($\rho = 0.57 \text{ g/cm}^3$). Wood density across species varies widely from 0.25 g/cm^3 (balsa) to $>0.90 \text{ g/cm}^3$ (ironwood/sal). While stand aggregation attenuates random variance, precise stand-level crediting requires ground-verified dominant species mapping.
- **Single-Epoch Snapshot vs. Multi-Temporal Additionality:** Clover currently evaluates a static temporal snapshot ($T_{\blacksquare}$). Verra credit issuance mandates multi-temporal diffing ($T_{\blacksquare} - T_{\blacksquare}$) across consecutive years to verify permanent sequestration and demonstrate additionality beyond business-as-usual baseline scenarios.

Summary: Clover is not a black-box wrapper. It is an honest, mathematically rigorous, and auditable remote sensing accounting pipeline designed from the ground up for transparent carbon project screening.